

Target-Rate Least-Squares Power Allocation over Parallel Channels

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Abstract—We study power allocation over N parallel Gaussian channels, such as OFDM subcarriers or the zero-forcing streams of a multiuser MIMO downlink, when each channel has a target spectral efficiency set by its receiver’s demand. We minimize the total squared deviation between achieved and target rates under a sum-power constraint. We prove that the optimum never overshoots a target, that the problem is equivalent to a strictly convex program, and that it has two regimes: if the budget suffices, every target is met exactly and the surplus power is left unused, unlike classical waterfilling; otherwise all of the available power is used and some targets are missed. In this power-limited regime, the optimum has a waterfilling-like closed form in which each channel has its own water level, given by the Lambert W function and proportional to the channel’s rate shortfall. The closed form extends to any ℓ_p shortfall penalty with $1 < p < \infty$, a family whose endpoints are capped waterfilling ($p = 1$) and minimax allocation ($p \rightarrow \infty$). The optimum is found by a one-dimensional search over a single dual variable; a compiled safeguarded Newton version converges in at most 6 iterations and runs 43 to 103 times faster than an interior-point solver for N up to 1024. On Rayleigh-faded channels at 5 dB mean SNR, the allocation serves 73.9% of demand versus 79.5% for capped waterfilling, while reducing the fraction of channels left without power from 10.1% to 3.9%.

Index Terms—Power allocation, waterfilling, parallel channels, multiuser MIMO, OFDM, Lambert W function, convex optimization.

I. INTRODUCTION

Waterfilling is the classical answer to allocating power over parallel Gaussian channels: it maximizes the sum of rates by pouring more power into stronger channels [1]–[3]. In multiuser systems, however, the parallel channels often serve *different receivers*. OFDMA subcarriers are assigned to different users, and zero-forcing precoding with unit-norm beams turns a multiuser MIMO downlink into parallel scalar channels, one per receiver stream, sharing a total power budget. Each receiver then has its own rate target derived from its demand. Rate beyond a target is an unnecessary waste of power; rate below it is demand left unmet, and how that deficit is spread across channels is how it is spread across receivers.

Existing formulations handle targets in two ways. Margin-adaptive loading treats them as hard constraints, minimizing transmit power [4] or maximizing the noise margin [5] at fixed target rates; see [6] for a survey. When the budget cannot meet every target, a choice is needed about whom to shortchange. Capped waterfilling, i.e., waterfilling with per-channel power

caps [7] set at the power that meets each target, maximizes the demand served, $\sum_i \min(r_i, T_i)$. It is efficient but places the whole deficit on the weakest channels, which receive no power at all. Fairness criteria such as proportional fairness [8], proportional rate constraints [9] and α -fairness [10] control rate *ratios*, not deviations from absolute targets.

We study a soft alternative: minimize the total squared deviation $\sum_i (\log_2(1+a_i P_i) - T_i)^2$ under a sum-power budget. The quadratic penalty grows with the size of a shortfall, so it spreads the deficit across receivers instead of concentrating it, and, like capped waterfilling, it never spends power above a target. Our contributions are:

- **Structure.** The optimum never overshoots; the problem is equivalent to a strictly convex program; and it has a sharp two-regime structure, leaving power unused whenever every target can be met (Section III).
- **Closed form.** In the power-limited regime, the optimum is $P_i = [L_i - 1/a_i]^+$, like waterfilling but with a channel-specific water level $L_i = 2W_0(\lambda \ln^2 2 q_i/2)/(\lambda \ln^2 2)$, where $q_i = 2^{T_i}/a_i$ and λ is a scalar dual variable; each level is proportional to the channel’s rate shortfall. The form extends to every ℓ_p shortfall penalty with $1 < p < \infty$, whose endpoints are capped waterfilling ($p = 1$) and minimax allocation ($p \rightarrow \infty$) (Section IV).
- **Algorithm.** The optimum is found by a one-dimensional search over the scalar dual variable, costing N Lambert- W evaluations per iteration; with a safeguarded Newton step it converges in at most 6 iterations (Section V).
- **Evaluation.** On Rayleigh-faded channels, the allocation gives up 2–6 points of served demand relative to capped waterfilling and leaves 2.6 – $4.1 \times$ fewer channels without power; capped waterfilling keeps the better 5th-percentile satisfaction at 10 dB (Section VI). A normalized weighting handles unequal targets (Remark 2).

Related work. Practical algorithms exist for generalized waterfilling with multiple water levels and multiple constraints [11], and mercury/waterfilling gives the optimal allocation for arbitrary input distributions [12]. The Lambert W function [13] yields closed-form power allocations in cooperative relay networks [14] and in energy-aware link adaptation [15]. Here it solves a different equation, the balance between a channel’s rate shortfall and its marginal rate. To our

TABLE I
NOTATION.

a_i, T_i	gain-to-noise ratio and target rate (b/s/Hz) of channel i
P_i, r_i	power and rate $\log_2(1 + a_i P_i)$ of channel i
P_{tot}	total power budget
$\bar{P}_i$	cap $(2^{T_i} - 1)/a_i$: power that meets target i exactly
q_i	$2^{T_i}/a_i = \bar{P}_i + 1/a_i$
δ_i	shortfall $T_i - r_i$
L_i	water level of channel i ; equals $P_i + 1/a_i$ when active
$\lambda, S(\lambda)$	dual variable; total power allocated at λ
c	$\ln 2$

knowledge, the squared target-deviation objective, its Lambert- W solution, and the resulting ℓ_p family have not been treated before.

II. SYSTEM MODEL AND PROBLEM

Consider N parallel Gaussian channels with gain-to-noise ratios $a_i > 0$, known at the transmitter. Power $P_i \geq 0$ on channel i yields spectral efficiency $r_i(P_i) = \log_2(1 + a_i P_i)$. Each channel has a target $T_i > 0$ (a channel with $T_i = 0$ receives no power and is dropped). We solve

$$\min_{\mathbf{P} \geq 0} J(\mathbf{P}) = \sum_{i=1}^N (r_i(P_i) - T_i)^2 \quad \text{s.t.} \quad \sum_{i=1}^N P_i \leq P_{\text{tot}}. \quad (1)$$

Two quantities recur. The cap $\bar{P}_i = (2^{T_i} - 1)/a_i$ is the power that meets target i exactly, and

$$q_i \triangleq \bar{P}_i + \frac{1}{a_i} = \frac{2^{T_i}}{a_i} \quad (2)$$

is the value of $P_i + 1/a_i$ at which channel i reaches its target. We write $\delta_i = T_i - r_i$ for the shortfall and $c = \ln 2$; a channel is *active* if $P_i > 0$. Table I collects the notation.

As an example of the model, consider a multiuser MIMO downlink with channel matrix $\mathbf{H}$ (one row per single-antenna receiver) and noise power σ^2 . Zero-forcing precoding with unit-norm beams gives each receiver an interference-free stream with $a_i = 1/(\sigma^2[(\mathbf{H}\mathbf{H}^H)^{-1}]_{ii})$, and the stream powers P_i add up to the transmit power.

III. STRUCTURE OF THE OPTIMUM

Lemma 1 (No overshoot). *Every optimal solution of (1) satisfies $P_i^* \leq \bar{P}_i$, i.e., $r_i(P_i^*) \leq T_i$, for all i .*

Proof. If $P_i > \bar{P}_i$, replacing P_i by $\bar{P}_i$ reduces $(r_i - T_i)^2$ to zero and keeps the budget feasible, strictly decreasing J . $\square$

Lemma 1 lets us restrict (1) to the box $\mathcal{B} = \{0 \leq P_i \leq \bar{P}_i\}$, on which $J(\mathbf{P}) = \sum_i \phi_i(P_i)$ with $\phi_i(P) = [(T_i - r_i(P))^+]^2$.

Proposition 1 (Convexity). *Each ϕ_i is convex on $[0, \infty)$ and strictly convex on $[0, \bar{P}_i]$, where*

$$\phi_i''(P) = \frac{2a_i^2}{c^2(1 + a_i P)^2} (1 + c\delta_i(P)) > 0. \quad (3)$$

Hence (1) is equivalent to the strictly convex program of minimizing $\sum_i \phi_i(P_i)$ over $\mathcal{B} \cap \{\sum_i P_i \leq P_{\text{tot}}\}$, and has a unique optimum.

Proof. $(T_i - r_i)^+$ is convex and nonnegative because r_i is concave, and $t \mapsto t^2$ is convex and nondecreasing on $t \geq 0$, so the composition ϕ_i is convex [16]. Differentiating twice on $[0, \bar{P}_i]$, with $r'_i = a_i/(c(1 + a_i P))$, gives (3). Strict convexity of the objective over a convex set gives uniqueness. $\square$

The unconstrained objective in (1) is not convex beyond $r_i = T_i + 1/c$; the no-overshoot property is what makes the problem tractable.

Theorem 1 (Two regimes). (A) Surplus. *If $P_{\text{tot}} \geq \sum_i \bar{P}_i$, then $\mathbf{P}^* = \bar{\mathbf{P}}$, $J^* = 0$, and $P_{\text{tot}} - \sum_i \bar{P}_i$ is left unused.* (B) Deficit. *If $P_{\text{tot}} < \sum_i \bar{P}_i$, then $J^* > 0$ and the budget is tight, $\sum_i P_i^* = P_{\text{tot}}$.*

Proof. (A) $\bar{\mathbf{P}}$ is feasible with $J = 0$, and by Lemma 1 and uniqueness it is the optimum. (B) Meeting every target needs $\sum_i \bar{P}_i > P_{\text{tot}}$, so $J^* > 0$ and some i has $P_i^* < \bar{P}_i$. Since $\phi'_i(P) = -2a_i\delta_i/(c(1 + a_i P)) < 0$ there, a slack budget would allow increasing P_i and lowering J . $\square$

In regime (A) the formulation departs most from waterfilling, which always spends the full budget; leaving the surplus unused saves transmit energy, in the spirit of using spare capacity to conserve terminal energy [17]. Capped waterfilling shares this property; the two differ in regime (B).

IV. CLOSED-FORM SOLUTION

Theorem 2 (Lambert- W solution). *In regime (B), the unique optimum has the waterfilling-like form*

$$P_i^* = \left[L_i(\lambda^*) - \frac{1}{a_i} \right]^+, \quad L_i(\lambda) = \frac{2}{\lambda c^2} W_0\left(\frac{\lambda c^2 q_i}{2}\right), \quad (4)$$

where W_0 is the principal branch of the Lambert W function, i.e., for $z \geq 0$, $W_0(z)$ is the unique $w \geq 0$ with $we^w = z$, and $\lambda^* > 0$ is the unique root of $S(\lambda) \triangleq \sum_i P_i(\lambda) = P_{\text{tot}}$.

Proof. The convex program of Proposition 1 satisfies Slater's condition, so KKT is necessary and sufficient. With multipliers λ for the budget, μ_i for $P_i \geq 0$ and ν_i for $P_i \leq \bar{P}_i$,

$$-\frac{2a_i\delta_i}{c(1 + a_i P_i)} + \lambda - \mu_i + \nu_i = 0. \quad (5)$$

Here $\lambda > 0$: with $\lambda = 0$, every channel with $P_i < \bar{P}_i$ would need $\delta_i = 0$, forcing $\mathbf{P} = \bar{\mathbf{P}}$, which is infeasible in (B). At $P_i = \bar{P}_i$, (5) gives $\nu_i = -\lambda < 0$, so no cap binds. *Active channels* ($0 < P_i < \bar{P}_i$, $\mu_i = \nu_i = 0$): substituting $1 + a_i P_i = 2^{T_i - \delta_i} = a_i q_i e^{-c\delta_i}$ into (5) gives

$$(c\delta_i) e^{c\delta_i} = \frac{\lambda c^2 q_i}{2}, \quad (6)$$

which has the form $we^w = z$, so its unique nonnegative root is $c\delta_i = W_0(\lambda c^2 q_i/2)$. Then

$$P_i + \frac{1}{a_i} = q_i e^{-c\delta_i} = q_i \frac{W_0(z_i)}{z_i} = L_i(\lambda), \quad z_i = \frac{\lambda c^2 q_i}{2}, \quad (7)$$

using $e^{-W_0(z)} = W_0(z)/z$. *Inactive channels* ($P_i = 0$, $\delta_i = T_i$): $\mu_i = \lambda - 2a_i T_i/c \geq 0$. Because W_0 is increasing and $W_0(cT_i e^{cT_i}) = cT_i$, the condition $\lambda \geq 2a_i T_i/c$ is equivalent to $W_0(\lambda c^2 q_i/2) \geq cT_i$, i.e., to $L_i(\lambda) = q_i e^{-W_0(\cdot)} \leq 1/a_i$; the $[\cdot]^+$ in (4) selects the right case. Since every $T_i > 0$, each q_i is finite and positive; $L_i(\lambda)$ is decreasing in λ because $W_0(z)/z$ is decreasing, so each $P_i(\lambda)$ is continuous and nonincreasing, strictly decreasing while active, with $S(0^+) = \sum_i \bar{P}_i > P_{\text{tot}}$ and $S(\lambda) = 0$ for $\lambda \geq \max_i 2a_i T_i/c$; hence λ^* is unique. $\square$

Waterfilling has the same form, $P_i = [\mu - 1/a_i]^+$, with one water level μ shared by all channels. Here each channel has its own level L_i , set by its $q_i = 2^{T_i}/a_i$. The proof also gives the shortfall directly.

Corollary 1 (Shortfall and water level). *At the optimum, every channel's shortfall is*

$$\delta_i(\lambda^*) = \min \left\{ T_i, \frac{1}{c} W_0 \left(\frac{\lambda^* c^2 q_i}{2} \right) \right\}, \quad (8)$$

and every active channel satisfies

$$P_i^* + \frac{1}{a_i} = L_i(\lambda^*) = \frac{2}{c\lambda^*} \delta_i. \quad (9)$$

The water level of each active channel is thus proportional to its own shortfall: channels with larger shortfalls sit higher. Equivalently, the optimum equalizes the shortfall-weighted marginal rate, $2\delta_i \partial r_i / \partial P_i = \lambda^*$.

For intuition, W_0 is increasing, $W_0(z) \approx z$ as $z \rightarrow 0$, and $W_0(z) = \ln z - \ln \ln z + o(1)$ as $z \rightarrow \infty$. When power is plentiful ($\lambda \rightarrow 0$), $\delta_i \approx \lambda c q_i/2$ and $L_i \rightarrow q_i$, recovering the caps; as power becomes scarce, shortfalls grow only logarithmically in λq_i until they reach T_i and the channel is switched off.

Corollary 2 (Ordering and activity). (i) *An active channel's shortfall depends on (a_i, T_i) only through $q_i = 2^{T_i}/a_i$ and increases with it: active channels that need more power to reach their target bear more of the deficit.* (ii) *Channel i receives zero power iff $a_i T_i \leq c\lambda^*/2$, so channels are switched off in increasing order of $a_i T_i$. Unlike waterfilling, a high target can keep a weak channel active.* (iii) *Every shortfall is nonincreasing in P_{tot} , and strictly decreasing for active channels: no channel loses from a larger budget.*

Proof. (i) and (ii) follow from (8) and the monotonicity of W_0 . For (iii), $S(\lambda)$ is decreasing, so λ^* decreases as P_{tot} grows, and each $\delta_i(\lambda)$ is nondecreasing in λ . $\square$

Remark 1 (The ℓ_p family). Replace the squared penalty by $J_p = \sum_i [(T_i - r_i)^+]^p$. For every $p \geq 1$, J_p is strictly convex on $\mathcal{B}$. For $p > 1$, repeating the proof with $m = p - 1$ gives $\delta_i^m e^{c\delta_i} = \lambda c q_i/p$, hence

$$\delta_i(\lambda) = \min \left\{ T_i, \frac{m}{c} W_0 \left(\frac{c}{m} \left(\frac{\lambda c q_i}{p} \right)^{1/m} \right) \right\}, \quad (10)$$

and again $P_i^* = [L_i - 1/a_i]^+$, now with level $L_i = p \delta_i^{p-1}/(c\lambda) \propto \delta_i^{p-1}$. At the endpoint $p = 1$, $J_1 = \sum_i T_i -$

Algorithm 1 Target-rate least-squares allocation

Require: gains a_i , targets $T_i > 0$, budget $P_{\text{tot}} > 0$, tolerance

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 $\varepsilon$ 
1:  $\bar{P}_i \leftarrow (2^{T_i} - 1)/a_i$ ,  $q_i \leftarrow 2^{T_i}/a_i$ 
2: if  $\sum_i \bar{P}_i \leq P_{\text{tot}}$  then return  $\mathbf{P}^* = \bar{\mathbf{P}}$   $\triangleright$  regime (A)
3:  $[\lambda_{\text{lo}}, \lambda_{\text{hi}}] \leftarrow$  bracket (11)
4: while  $\lambda_{\text{hi}}/\lambda_{\text{lo}} > 1 + \varepsilon$  do
5:    $\lambda \leftarrow \sqrt{\lambda_{\text{lo}} \lambda_{\text{hi}}}$   $\triangleright$  bisection on  $\log \lambda$ 
6:    $P_i \leftarrow [L_i(\lambda) - 1/a_i]^+$  via (4), for all  $i$ 
7:   if  $\sum_i P_i > P_{\text{tot}}$  then  $\lambda_{\text{lo}} \leftarrow \lambda$  else  $\lambda_{\text{hi}} \leftarrow \lambda$ 
8: return  $P_i^* = [L_i(\lambda_{\text{hi}}) - 1/a_i]^+$ 

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$\sum_i \min(r_i, T_i)$ on $\mathcal{B}$, and the KKT conditions give a common level $P_i + 1/a_i = \mu$ capped at q_i : capped waterfilling. As $p \rightarrow \infty$ the objective approaches minimax shortfall, $\min \max_i \delta_i$. The exponent p is thus a single knob from served demand to equity, with the closed form (10) at every interior point.

Remark 2 (Extensions). Weights $w_i > 0$ in J replace λ by λ/w_i in (4) and (8), and the bracket of Algorithm 1 becomes $\lambda_{\text{lo}} = (\sum_i \bar{P}_i - P_{\text{tot}})/(c^2 \sum_i q_i^2/w_i)$, $\lambda_{\text{hi}} = \max_i 2w_i a_i T_i/c$. With unequal targets, the unweighted objective drops low-target channels first (Corollary 2(ii)); the weights $w_i = T_i^{-2}$, which penalize relative rather than absolute shortfall, correct this. With T_i drawn uniformly from $\{1, 2, 4\}$ b/s/Hz in the Rayleigh setup of Section VI at 10 dB, the weighted allocation reaches a mean satisfaction $\min(r_i, T_i)/T_i$ of 90.8% (capped waterfilling: 92.9%) and leaves 0.45% of channels without power (2.20%). A coding scheme with SNR gap Γ , $r_i = \log_2(1 + a_i P_i/\Gamma)$, only rescales $a_i \rightarrow a_i/\Gamma$. With separate power budgets for disjoint groups of channels, the problem separates across groups, and each is solved by (4).

V. ALGORITHM

Theorem 2 reduces the problem to finding the root of the monotone scalar function $S(\lambda)$. Bisection needs a starting interval, or *bracket*, that is known to contain λ^* . Such an interval is usually found by trial, for example by doubling a guess until S crosses P_{tot} . The following result gives both endpoints as formulas, so no trial is needed.

Proposition 2 (Explicit bracket). *In regime (B), $\lambda^* \in [\lambda_{\text{lo}}, \lambda_{\text{hi}}]$ with*

$$\lambda_{\text{lo}} = \frac{\sum_i \bar{P}_i - P_{\text{tot}}}{c^2 \sum_i q_i^2}, \quad \lambda_{\text{hi}} = \max_i \frac{2a_i T_i}{c}. \quad (11)$$

Proof. By the inactivity condition in the proof of Theorem 2, every channel is off at λ_{hi} , so $S(\lambda_{\text{hi}}) = 0 < P_{\text{tot}}$ (assuming $P_{\text{tot}} > 0$). For the lower end, with $\delta_i(\lambda)$ as in (8), $W_0(z) \leq z$ gives $\delta_i \leq \lambda c q_i/2$, and $1 - 2^{-\delta} \leq c\delta$ gives $P_i(\lambda) = \bar{P}_i - q_i(1 - 2^{-\delta_i}) \geq \bar{P}_i - \lambda c^2 q_i^2/2$. Summing, $S(\lambda_{\text{lo}}) \geq \sum_i \bar{P}_i - \frac{1}{2}(\sum_i \bar{P}_i - P_{\text{tot}}) > P_{\text{tot}}$. $\square$

Each iteration costs N evaluations of W_0 , which Halley's iteration computes to machine precision in a few steps [13]. Bisection on $\log \lambda$ needs $\lceil \log_2(\ln(\lambda_{\text{hi}}/\lambda_{\text{lo}})/\varepsilon) \rceil$ iterations, so the total cost is $O(N(\log(1/\varepsilon) + \log \log(\lambda_{\text{hi}}/\lambda_{\text{lo}})))$; the second term grows only as P_{tot} approaches $\sum_i \bar{P}_i$.

Newton variant. With $u = \ln \lambda$ and $\omega_i = W_0(\lambda c^2 q_i/2)$,

$$\frac{dS}{du} = - \sum_{i \text{ active}} \frac{(1 + a_i P_i) \omega_i}{a_i (1 + \omega_i)}, \quad (12)$$

which is available in closed form between the points where channels switch off, where S has kinks. We therefore take a Newton step in u , fall back to bisection whenever the step leaves the current bracket, update the bracket at every iteration, and stop when $|S(\lambda) - P_{\text{tot}}| \leq \varepsilon P_{\text{tot}}$.

VI. NUMERICAL RESULTS

We compare against five baselines, each with its own optimality criterion: *waterfilling* (max $\sum_i r_i$); *capped waterfilling* (max served demand $\sum_i \min(r_i, T_i)$, the $p = 1$ endpoint of Remark 1); *max-min satisfaction* (max $\min_i r_i/T_i$, which with equal targets is the $p \rightarrow \infty$ endpoint); *proportional fairness* (max $\sum_i \log r_i$); and *uniform power*. We checked Algorithm 1, the ℓ_p form (10) (p from 1.25 to 10) and the weighted form against CVXPY/Clarabel on more than 850 random instances; all relative objective gaps are below 10^{-7} , the solver's tolerance. SLSQP agrees on every instance where it was run. Code and data are available at <https://github.com/ANRGUSC/target-rate>.

A. Worked example

Fig. 1 uses $N = 8$ channels with gains $\{20, 15, 10, 7, 5, 3, 2, 1\}$, $T = 3$ b/s/Hz and $P_{\text{tot}} = 10 < \sum_i \bar{P}_i = 16.75$ (regime B). Waterfilling overshoots the four strongest channels by up to 1.95 b/s/Hz and leaves the weakest at 0.63. Capped waterfilling meets five targets exactly and pushes the deficit onto the three weakest channels (rates 2.98, 2.40 and 1.40). The target-rate allocation keeps every channel below but near its target, with shortfalls from 0.09 to 0.98 b/s/Hz ordered by q_i (Corollary 2); its deviation is $J^* = 1.79$, versus 2.92 for capped waterfilling, 3.15 for max-min and 15.4 for waterfilling. Here $\lambda^* = 0.70$. The strongest channel ($a_i = 20$, $q_i = 0.40$) has level $L_i = 0.375$, power 0.325 and shortfall 0.09; the weakest ($a_i = 1$, $q_i = 8$) has level 4.05, power 3.05 and shortfall 0.98. In both, the level equals $2\delta_i/(c\lambda^*)$, as in (9). Fig. 1(b) shows the levels $P_i^* + 1/a_i$ of (9), which rise with the shortfall, next to the flat waterfilling level.

Fig. 2 sweeps the budget. Below $\sum_i \bar{P}_i = 16.75$, all methods spend the whole budget and the target-rate allocation has the smallest deviation J . Above it, the allocation reaches $J = 0$ and stops spending: at $P_{\text{tot}} = 25$ it uses 67% of the budget, whereas waterfilling and uniform allocation spend all of it and *increase* J through overshoot. With heterogeneous targets $\{5, 4, 3, 3, 2, 2, 1, 1\}$ ($\sum_i \bar{P}_i = 7.35$), a budget of 15 leaves 7.65 units unused with every target met.

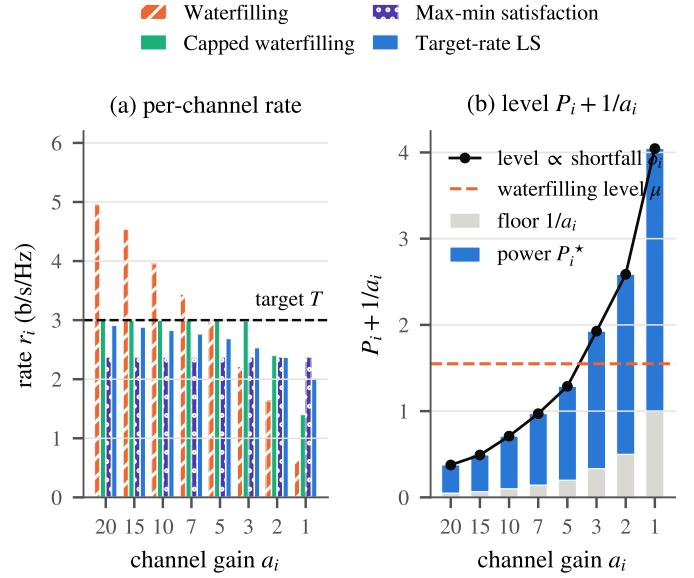


Fig. 1. Worked example ($N = 8$, $T = 3$, $P_{\text{tot}} = 10$). (a) Achieved rates. (b) Target-rate levels $P_i^* + 1/a_i$: the grey bar is the floor $1/a_i$, the blue bar the allocated power, and the black line the resulting level, which is proportional to each channel's shortfall. The dashed line is the single level of waterfilling.

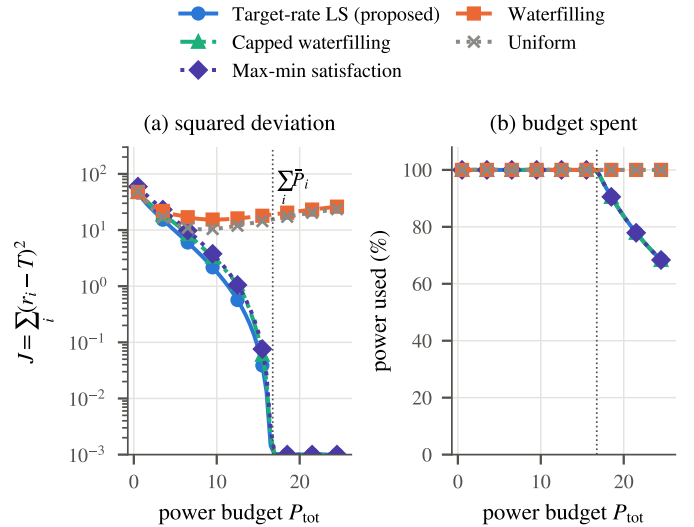


Fig. 2. Budget sweep for the example of Fig. 1. The dotted line marks $\sum_i \bar{P}_i$, the transition from regime (B) to regime (A). Capped waterfilling and max-min also stop at the caps, so these three curves coincide in (b).

B. Rayleigh-faded channels

We next draw $N = 64$ channels with $a_i = \bar{\gamma} |h_i|^2$, $|h_i|^2 \sim \text{Exp}(1)$ i.i.d., set $P_{\text{tot}} = N$ so that $\bar{\gamma}$ is the mean SNR, use a common target $T = 2$ b/s/Hz, and average 500 realizations per SNR. We report the *satisfaction* $\min(r_i, T)/T$ of each channel. Its mean is the fraction of demand served, which capped waterfilling maximizes by construction; its lower tail (5th percentile, P5) and the fraction of *starved* (zero-power) channels capture how the deficit is shared among receivers. We

TABLE II
RAYLEIGH-FADED CHANNELS ($N = 64$, $T = 2$). MEAN AND P5: MEAN AND 5TH PERCENTILE OF THE SATISFACTION $\min(r_i, T)/T$ (%). STARVED: CHANNELS WITH ZERO POWER (%). DEV.: MEAN OF $(r_i - T)^2$. BOLD: BEST; UNDERLINED: SECOND BEST.

Method	$\bar{\gamma} = 5$ dB				$\bar{\gamma} = 10$ dB			
	Mean	P5	Starved	Dev.	Mean	P5	Starved	Dev.
Target-rate	73.9	18.4	3.87	0.51	96.2	85.4	0.18	0.047
Capped WF	79.5	1.0	10.08	<u>0.65</u>	97.8	90.4	0.74	0.064
Max-min	32.1	32.1	0.00	1.92	73.6	73.6	0.00	0.51
Prop. fair	72.8	17.7	0.00	0.77	92.2	49.9	0.00	1.79
Waterfilling	67.7	0.0	16.93	1.80	85.1	6.6	7.03	3.43
Uniform	71.7	13.0	0.00	1.04	89.5	36.5	0.00	2.56

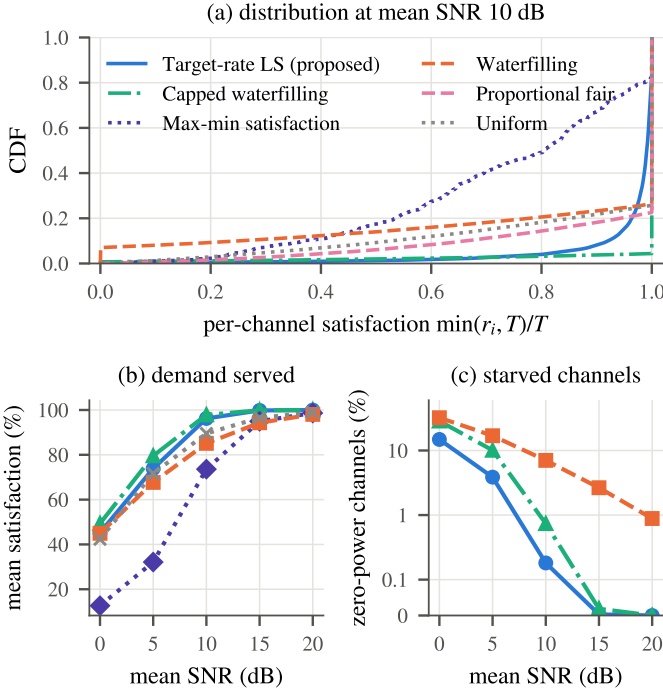


Fig. 3. Rayleigh-faded channels. (a) CDF of per-channel satisfaction at 10 dB. (b) Mean satisfaction (served demand). (c) Fraction of channels given zero power (symmetric-log axis); max-min, proportional fairness and uniform allocation starve none.

also report the mean squared deviation $(r_i - T)^2$, the criterion the target-rate allocation minimizes. Each metric is computed over the 64 channels of one realization (P5 is the 5th percentile across those channels) and then averaged over realizations. Table II and Fig. 3 summarize.

Each target-aware method is best on its own criterion: capped waterfilling on served demand, max-min on the worst-served channel, and the target-rate allocation on squared deviation. The question is what each gives up on the others' criteria. Three observations stand out. First, the target-rate allocation gives up little efficiency: it serves 73.9% of demand at 5 dB and 96.2% at 10 dB, versus 79.5% and 97.8% for capped waterfilling, whereas max-min gives up more than 47 points at 5 dB. Second, it spreads the deficit rather than concentrating it.

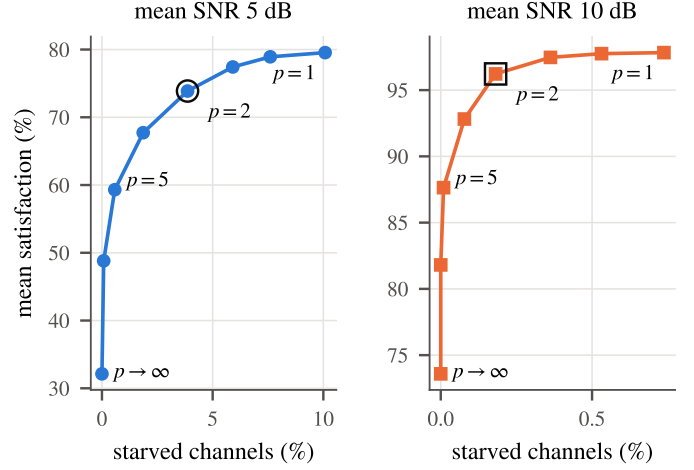


Fig. 4. Served demand versus starved channels across the ℓ_p shortfall family, $p \in \{1, 1.25, 1.5, 2, 3, 5, 10, \infty\}$. The ringed point is $p = 2$.

At 5 dB, starved channels fall from 10.1% to 3.9% and the 5th-percentile satisfaction rises from 1.0% to 18.4%. This benefit is largest when power is scarce and shrinks as power becomes plentiful: at 10 dB, where only a few deeply faded channels are at risk, capped waterfilling is ahead on both mean and 5th percentile (90.4% vs. 85.4%), although it still starves $4.1\times$ more channels (0.74% vs. 0.18%). Third, the allocations that never starve behave differently. Max-min serves only 32.1% at 5 dB. Proportional fairness nearly matches the target-rate allocation at 5 dB (72.8% served, P5 17.7%, no starvation), but drops to 92.2% served and a 49.9% P5 at 10 dB, because it keeps spending power above target on strong channels. From 15 dB on, regime (A) dominates (84% of realizations at 15 dB, 96% at 20 dB): the target-rate allocation meets essentially all demand using 57% and 23% of the budget, while waterfilling, proportional fairness and uniform allocation spend all of it and still serve less.

C. The ℓ_p trade-off

Fig. 4 traces the ℓ_p family of Remark 1 on the equal-target channels; every point with $1 < p < \infty$ comes from the closed form (10), and the endpoints from capped waterfilling and minimax bisection. Raising p trades mean satisfaction for fewer starved channels. At 5 dB, moving from $p = 1$ to $p = 2$ costs 5.7 points of served demand and removes 62% of starved channels; continuing to $p = 5$ removes most of the rest but costs another 14.6 points. At 10 dB, $p = 2$ gives up 1.6 points and cuts starvation by 75%. The least-squares objective thus sits near the knee in both regimes, and (10) offers the other operating points at the same cost.

D. Computation time

Table III reports median times over 10 random regime-(B) instances per N , all measured on the same machine. Algorithm 1 is timed compiled (Numba) with bisection ($\varepsilon = 10^{-12}$,

TABLE III
MEDIAN SOLVE TIME (MS) AND SPEEDUP OF THE NEWTON VARIANT OVER CLARABEL (REGIME B, $a_i = 10 |h_i|^2$, $T = 2$, $P_{\text{tot}} = \min(N, \frac{1}{2} \sum_i \bar{P}_i)$).

N	Algorithm 1			Reference solvers		Speedup
	Newton	Bisect.	NumPy	Clarabel	SLSQP	
8	0.007	0.038	0.45	0.30	2.01	$43\times$
64	0.043	0.26	1.03	2.30	17.9	$54\times$
256	0.17	1.03	2.63	12.1	1,006	$73\times$
1,024	0.66	4.05	8.70	68.1	49,293	$103\times$

43–44 iterations) and with the safeguarded Newton step (4–6 iterations), and as vectorized NumPy. For the Clarabel interior-point solver, the CVXPY problem is compiled once and we count only the solve time Clarabel reports; SLSQP uses analytic gradients. The Newton variant is $43\text{--}103\times$ faster than Clarabel’s solve time alone, the gap widening with N , and the two agree to a relative objective of 1.3×10^{-9} . At $N = 4096$, Newton takes 2.7 ms, while Clarabel failed on one of three instances and returned an inaccurate solution on another. A practical tolerance $\varepsilon = 10^{-6}$ cuts bisection to 23–24 iterations with objective within 10^{-6} (relative) of the optimum.

E. Scope

The results assume that the gains a_i are known exactly at the transmitter, that rates follow the Shannon formula, and that any rate up to the target is useful. With estimated gains, the no-overshoot property holds only with respect to the estimates. With a discrete set of rates, a channel left slightly below its target would have to step down to the next available rate, so the allocation would need a rounding stage. We do not evaluate either case here.

VII. CONCLUSION

Minimizing the squared deviation from per-channel rate targets gives an allocation that never overshoots, returns surplus power, and has a closed form: the optimal shortfall of each channel is a Lambert- W function of $2^{T_i}/a_i$ and a scalar dual variable. The same form covers every ℓ_p penalty with $1 < p < \infty$, placing capped waterfilling and minimax allocation at the two ends of one family. The optimum costs a few Lambert- W evaluations per channel. Open directions include imperfect channel knowledge, discrete rate sets, finite-blocklength rate models, and joint channel assignment with the closed form as the inner solver.

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